

Review

A Review on PM_{2.5} Sources, Mass Prediction, and Association Analysis: Research Opportunities and Challenges

Peng-Yeng Yin 

Information Technology and Management Program, Ming Chuan University, No. 5 De-Ming Road,
Gui-Shan District, Taoyuan City 333321, Taiwan; pyyin@mail.mcu.edu.tw

Abstract: Air pollution has long been one of the most life-threatening issues which has led to massive amounts of premature human death due to fatal diseases and environmental disasters. Several Sustainable Development Goals (SDGs) set up by the United Nations coincide with the solutions for air pollution reduction. To reach a sustainable future, researchers have conducted many theoretical analyses or case studies of air pollution at different places on the globe and proposed prudent strategies for obtaining an equilibrium between socioeconomic development and air pollution reduction. This research selected a substantial number of articles and existing review papers published between 2013 and 2024 and organized these publications into subfields. This research was focused on filling the gap between existing reviews and the state-of-the-art technologies in the last decade. To be informative and contextual, this review presented a pathway for readers to comprehend the research in three contiguous phases of air pollution analysis, from compositional apportionment and mass prediction of pollution to disclosing associations between pollution concentration and natural or anthropogenic factors. At the end of this review, the author highlighted several research fields which have been overlooked in previous reviews but will be increasingly important in the future.

Keywords: PM_{2.5}; air pollution; Sustainable Development Goals; source apportionment; pollution mass prediction; association analysis



Academic Editor: Konstantinos
Dimitriou

Received: 17 January 2025

Revised: 26 January 2025

Accepted: 27 January 2025

Published: 29 January 2025

Citation: Yin, P.-Y. A Review on PM_{2.5} Sources, Mass Prediction, and Association Analysis: Research Opportunities and Challenges. *Sustainability* **2025**, *17*, 1101. <https://doi.org/10.3390/su17031101>

Copyright: © 2025 by the author.
Licensee MDPI, Basel, Switzerland.
This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Air pollution is one of the major obstacles obstructing human beings from moving forward toward a sustainable future. Severe air pollution can cause human diseases [1], plant species mortality [2], atmospheric visibility deterioration [3], and global warming [4]. Carbon reduction or zero carbon has been the focus of global actions to mitigate the impact of air pollution. The country members of United Nations Framework Convention on Climate Change (UNFCCC) have signed several important action agreements at the meetings, including the Kyoto Protocol in 1997, the Paris Agreement in 2015, and the COP27 agreement in 2022. The United Nations also set up the Sustainable Development Goals (SDGs) to urge all nations to take actions for reducing air pollution. The relevant SDGs include good health and well-being (Goal 3); affordable and clean energy (Goal 7); decent work and economic growth (Goal 8); industry, innovation and infrastructure (Goal 9); sustainable cities and communities (Goal 11); responsible consumption and production (Goal 12); and climate action (Goal 13).

The ambient particulate matter with an aerodynamic diameter of less than 2.5 µm (PM_{2.5}) is the most threatening air pollutant which has been shown in the literature as having strong relationships with unexpected human deaths. It has been reported by the

World Health Organization (WHO) that ambient PM_{2.5} emissions were estimated to cause 4.2 million premature deaths worldwide in 2019 [5]. PM_{2.5} is a collection of complex compositions and comes from three major kinds of sources [6]. The natural sources include soiling, blown sea salt, volcanic eruptions, biological excrements, etc. The anthropogenic sources are those from socioeconomic activities such as industrial production, vehicle exhaust gas, and coal and gasoline combustion. The last kind of source is caused by the photochemical transformation of precursors such as SO₂ and NO_x. The concentration, transportation, and dispersion of PM_{2.5} are hard to analyze and predict due to dynamic and uncertain natural and anthropogenic activities and meteorological variables such as wind speed and direction, precipitation, temperature, relative humidity, atmospheric pressure, and solar radiation.

To analyze air pollution, appropriate sensors for measuring the pollution mass need to be deployed at air quality monitoring stations. There are two types of monitoring stations: supersites and microsites. The supersites are supported by governments and are high-cost and usually deployed only in populated cities or industrial districts. The typical way of measuring PM_{2.5} concentrations at supersites is to use the gravimetric method [7], which collects pollutant samples on a filter and measures the weight of pollutants with a designated size. Another alternative is the beta attenuation method [8], which emits carbon-14 radiation to pass through the filter with and without samples. The difference in beta ray decrement due to the absorption of suspended particles is used to calculate the particle mass.

On the contrary, microsites facilitated by low-cost sensors (LCSs) can be deployed at a denser distribution than supersites. However, LCSs are not highly accurate and thus microsites need to be calibrated with reference to supersites [9]. The World Meteorological Organization (WMO) suggested some principles for appropriate uses of LCSs (<https://wmo.int/publication-series/> accessed on 26 January 2025), including calibration, performance assessment, and data communication. The integration of LCS measurements with other data sources can improve data reliability and provide supportive evidence for observed air quality trends. The laser light scattering (LLS) method and quartz crystal microbalance (QCM) are the most prevailing techniques for measuring the PM_{2.5} concentrations at microsites [10]. The LLS method uses a laser beam to illuminate the suspended air particles to generate light scattering. The scattered light is detected by a photodiode detector at a certain angle to estimate the particle size and the number of particles. In QCM, the suspended particles concentrated on the oscillator will alter the mass and the frequency of the quartz crystal, which are then used to estimate the pollutant mass density.

In addition to ground sensors, satellite sensors can offer a broader monitoring area with various scales. Satellite sensors measure the aerosol optical depth (AOD), which reflects the ratio of light that is blocked by aerosols from penetrating the atmosphere. Ground-level PM_{2.5} concentrations can be estimated by the retrieved AOD data using spatiotemporal regression methods [11,12]. To reduce the estimation bias, local variables such as meteorological and land factors can be included as independent variables in the regression model.

This work conducted a retrospective review of PM_{2.5} source apportionment, mass prediction, and association analysis. The research of source apportionment is to identify the chemical compositions of air pollutants. The composition elements and their mass quantities can be used as trace markers to disclose potential emission sources and transportation paths of the pollution. The research of pollution mass prediction is to forecast the pollution concentrations at a time-ahead window by analyzing the trend of historical data. There exist both short-term and longer term look-ahead pollution mass predictions which forecast the future pollution concentration in a few minutes, hours, days, months,

or years. The third main aspect of air pollution analysis research is finding associations between pollution mass concentrations and related factors, ranging from diseases, festivals, land uses and landmarks, to solar radiation, volcanic eruptions, and wildfires. The reason these three research issues were chosen is because they are comprehensive in conducting air pollution analyses. The analysis starts with identifying the compositional elements of pollution and finding the relationship between the pollution apportionment and local natural or anthropogenic activities. The next stage will be predicting future concentrations of pollution from historical data of the elements or factors that resulted in the pollution. Finally, the last stage is to disclose what consequences in nature or human society may be expected if the predicted pollution concentration levels happen. This review was shaped to be informative and contextual based on the three research topics supported by related articles.

The three noted research problems are highly perplexing because the influential natural and anthropogenic factors which cause air pollution are dependent on locations and time periods. This phenomenon has resulted in many case studies of air pollution, and it is not uncommon to see inconsistent or even opposite conclusions from different case study articles. It was found that there was at least one review article published [6–14] in almost every year in last decade from 2014 to 2023, validating that the number of research papers on air pollution analysis is overwhelming. Attributed to the same reason as the huge volume of published papers, most of the existing reviews focused their discussions on only a single research topic, and some research topics were apparently overlooked. The motivation of this review is to fill the gap between existing reviews, the overlooked works, and the state-of-the-art technologies. In addition to the three noted research topics, the review also identified promising research opportunities and adherent challenges which will be increasingly important at the next stage of air pollution analysis research.

2. Methodologies

Air pollution analysis spans many research fields and there exist a great number of articles published in each field. To shape this review to be informative and contrasting with existing ones, the author first characterized the features and focuses of existing reviews. The author used keywords combined from the following terms: ‘air pollution’, ‘PM_{2.5}’, ‘review’, and ‘survey’. The nine most relevant reviews were identified and compared. The focused topics, the covered publication years of the literature, and the investigated region of the pollution datasets for the nine reviews are summarized in Table 1. Firstly, it is seen that in almost every year between 2014 and 2023 there was a published review related to air pollution, except for the years 2018–2020. This indicates that the number of published articles in relevant fields of air pollution grew rapidly in the time period, with the need for a new review sustained in every past year and the years to come. This is a motivation for this work to conduct a new review. Secondly, the topics discussed in each review were very focused. Five out of the nine reviews were focused on the impacts of air pollution on health [13,15–18]. The review by Mohammed et al. [14] discussed the literature which modeled the infiltration rate from outdoor pollutants to indoor ones. The review by Singh et al. [6] was specialized in source apportionment studies by the receptor model. Surprisingly, though air pollution prediction is one of the topics with the most public concern and there exist many published articles in this field, there is only one review by Dass et al. [19] conducted on this topic. Their review emphasized the prediction of vehicle CO emission, which is a typical pollution scenario in megacities in India. Instead of focusing on just one topic, the review by Liang et al. [20] gave comprehensive discussions in several fields, including chemical composition, source apportionment, control policies, and ideologies. Thirdly, for the range of publication years of the covered liter-

ature, some reviews [13,17,19,20] selected works from the literature published within a particular one or two years, while the others [6,14–16,18] chose works from the literature published in a broad time period. Lastly, for the region that the pollution dataset addressed, four reviews focused on particular cities, such as Shiraz [13]; cities in South Asia [6]; Bangalore, Delhi, and Mumbai [19]; and Tehran [17], while the others had no restriction on regions [14–16,18,20]. Considering the comparisons among the nine reviews, the author set the literature selection criteria as follows: The review focuses on the topics of source apportionment, PM_{2.5} mass prediction, and association analysis. The covered publication years are in the last decade between 2013 and 2024. There is no restriction on the investigated region of the selected literature. The author also identified several promising and challenging topics, namely, spatiotemporal analysis, maintenance of monitoring stations, pollution control policies, and ideologies, which emerged to be increasingly important after the last decade.

Table 1. Features of the nine reviews published between 2014 and 2023.

Published Reviews	Focused Topics	Covered Years	Regions
Dehghani et al., 2014 [13]	Impacts on health (cardiopulmonary mortality)	2011–2012	Shiraz, Iran
Mohammed et al., 2015 [14]	Outdoor–indoor infiltration	2003–2013	Worldwide
Liang et al., 2016 [20]	Chemical composition, source apportionment, control policy, and ideology	2013–2015	Worldwide
Singh et al., 2017 [6]	Source apportionment	2007–2016	South Asia
Dominski et al., 2021 [15]	Impacts on health (respiratory diseases)	Beginning~2020	Worldwide
Dass et al., 2021 [19]	Vehicle CO prediction	2016–2017	Bangalore, Delhi, and Mumbai, India
Li et al., 2022 [16]	Impacts on health (brain damage)	2004–2022	Worldwide
Hashemi et al., 2022 [17]	Impacts on health (COVID-19)	2020–2021	Tehran, Iran
Krittawong et al., 2023 [18]	Impacts on health (cardiovascular diseases)	1995–2023	Worldwide

The articles perused in this review were sought from Scopus (www.elsevier.com/solutions/scopus accessed on 26 January 2025) which fully covers Web of Science, MEDLINE, EMBASE, and ScienceDirect. To focus on the most recent and cutting-edge literature on the selected topic of air pollution, this review constrained the search criteria to journal papers published between 2013 and 2024 with the keywords combined from every term in the following two groups: Group A: air pollution, PM_{2.5}; Group B: source apportionment, predict, forecast, association, spatiotemporal, maintenance, policy, and ideology.

3. Source Apportionment and Emission Sources

Analysis of the chemical compositions of the air pollutants can provide informative clues for identifying the polluting emission sources and the major socioeconomic events which contribute to pollution. This helps to clarify the categories of chemical species causing human diseases, plant fatalities, reducing atmospheric visibility, etc. Among others, sulfate–nitrate–ammonium (SNA) water-soluble ions and polycyclic aromatic hydrocarbons (PAHs) are the main contributors to human respiratory diseases and cancers [21]. SNA, organic carbon (OC), and elemental carbon (EC) promote haze formation and degrade atmospheric visibility. Extreme precipitation and temperature are more often observed due to the growth of organic compounds which affect the nucleation of cloud droplets. There are three prevailing models for source apportionment analyses [22], i.e., the receptor model, the source model, and the tracer model. The appropriateness of the selected

source apportionment model depends on the purpose of the underlying application. Some remarkable applications for each model are reviewed as follows.

The receptor model measures the chemical composition of particulates at selected places and quantifies the contributions of emissions from various source categories at different places. According to the Chemical Mass Balance (CMB) principle, the mass change of the chosen marker species is calculated and the relationship between the marker species from the observed site and the emission sources can be resolved and interpreted by statistical techniques such as positive matrix factorization (PMF) and principal component analysis (PCA). With sufficient data measurement over time, the back trajectories of emission transportation are derived. A case study of PAHs in Changhua County in the central Taiwan area was conducted with 22 PAH species samples collected from 2014 to 2015 [21]. Surrounding Changhua County are located several large-scale coal-fired power plants and petrochemical complexes, imposing potential threats to human health. The PMF receptor-based apportionment model is applied to analyze the PAH compounds. The total PAHs showed a negative correlation with temperature and relative humidity, but a positive correlation with atmospheric pressure. Three main emission sources were identified, including unburned petroleum and traffic emissions (42%), steel industry and coal combustion (31%), and petroleum and oil burning (27%). The lifetime excess cancer risk (ECR) from inhalation exposure to PAHs is 4.7×10^{-5} , which is higher than the guideline limit (10^{-6}) released by the US EPA. Han et al. [23] applied the PMF model to apportion the PM_{2.5} chemical compositions in Baton Rouge, Louisiana, using six-year data from 2009 to 2014. Seven sources (OC, SO₄²⁻, S, NH₄⁺, NO₃⁺, EC, and Na) were identified and the top three sources were secondary sulfate, industry emissions, and secondary nitrate. The contributions of industry emissions and crustal dust have been increasing, while the contributions of traffic and sea salt decreased in these years. Secondary sources were higher than primary sources in the winter, while the crustal and road dust were dominant in the summer. It is evidenced in this study that winds play a significant role in the contributions of different sources in different seasons. Zhang et al. [24] harmonized the receptor model and the source-oriented model for the source categories of the hourly measured PM_{2.5} in Beijing between 17 January and 20 February 2021. The receptor model identified five sources, i.e., biomass burning, coal combustion, industrial emission, traffic emission, and fireworks, though fireworks only contributed about 4% of the total PM_{2.5} concentrations. The source-oriented model identified three major categories sources, i.e., residential combustion, traffic, and industry.

The source model obtains the chemical composition of emissions from samples and then extracts interesting species for analysis. Sufficient high-quality data are required to conduct numerical transportation and dispersion simulations on the trace species for identifying the anthropogenic activities which most contribute to the extracted chemical elements. Moreda-Pineiro et al. [25] analyzed eighteen samples of trace metals and inorganic ions at a northwest Atlantic European coastal suburban site during a 1.5 year period from March 2011 to August 2012. Their analysis showed that marine, crustal, and anthropogenic sources were the main contributors to the inorganic composition of PM_{2.5} and that the cleaner Atlantic air masses have a great influence on the emission reduction at the studied site. Ratanavalachai and Trivitayanurak [26] analyzed the PM_{2.5} concentration in downtown Bangkok which was contributed from four types of vehicles, including personal cars, light-duty vehicles (vans and pick-ups), heavy-duty vehicles (buses and trucks), and others (motorized and three-wheelers). The source model was applied with observatory PM_{2.5} data and traffic volume data of the four types of vehicles. The dispersion simulations showed that the major contributors to the PM_{2.5} concentration in downtown Bangkok were the traffic emissions from expressways (32%), followed by urban roads

(17.5%). The expressway emissions were mainly from heavy-duty vehicles, while the urban road emissions were evenly contributed by light-duty and heavy-duty vehicles. Fakhri et al. [27] conducted source apportionment of PM_{2.5} at an urban site in Montréal, Canada, during August–November 2020. The PMF tool was applied on chemical composition data. The authors used both secondary inorganic aerosols (SIA) and secondary organic aerosols (SOA) as molecular markers to examine sources and identified that SIA and SOA were the main components contributing 39 % of the PM_{2.5} mass, while another 23% of the PM_{2.5} mass was attributed to local anthropogenic sources, including traffic, road dust, industrial, and cooking emissions. The industrial emissions were minor contributors to total PM_{2.5} mass but were found to be the main sources for Co and Cr, which may cause health hazards like cancers.

Yin et al. [28] applied a specific tracer model method called the particulate source apportionment technology (PSAT) to simulate the concentrations of PM_{2.5} chemical components (SO_4^{2-} , NH_4^+ , NO_3^- , OC, and EC) contributed from eight emission source categories on areas in Guangdong province (GD), China. The data in 2012 were analyzed and the numerical simulations showed that the top three sources were industry sources (22.5%), mobile sources (19.3%), and agriculture sources (17.5%) which contributed to the PM_{2.5} emissions in GD. In particular, mobile sources contributed the most to the pollution observed in central GD, while industry sources were the main contributors to the PM_{2.5} in northeastern and southwestern regions in GD. Liu et al. [29] analyzed the apportionment of PM_{2.5} carbonaceous aerosols in Guangzhou, China by using radiocarbon and SOA tracers. They identified that the contributions of fossil carbon to EC, water-insoluble organic carbon (WIOC), and water-soluble organic carbon (WSOC) were $71 \pm 10\%$, $40 \pm 6\%$, and $33 \pm 3\%$. The main contributor of fossil carbon was vehicle exhaust.

Liao et al. [30] combined all three source apportionment models to analyze different source contributors to trace 8 metals, elemental carbon, primary organic carbon, and 10 other primary species in PM_{2.5}. The authors aimed at developing a more accurate protocol for the hybrid model source apportionment of trace metals. The source emissions data in the PRD region of China during 2015 were resolved and the result showed that the major contributors to Fe were area sources (41%), power generation (26%), and mobile emissions (26%). Cu was mainly contributed from area sources (31%), the industry sector (27%), and power generation (20%), while the emissions contributing to most of the quantities of Mn were from area sources (40%), outside PRD (23%), and power generation (17%).

Table 2 summarizes the used models, compositional elements, and emission sources of the reviewed articles.

Table 2. Summary of reviewed articles for source apportionment and emission sources.

Reference City Country	Model	Source Apportionment	Emission Sources
[21] Changhua Taiwan	receptor model	22 PAHs	unburned petroleum and traffic emissions (42%), steel industry and coal combustion (31%), and petroleum and oil burning (27%)
[23] Louisiana USA	receptor model	OC, SO_4^{2-} , S, NH_4^+ , NO_3^+ , EC, Na	secondary sulfate (38.4%), industry emissions (18.7%), secondary nitrate (17.6%), traffic (11.5%), crustal dust (6.1%), road dust (4.2%), and sea salt (3.6)
[24] Beijing China	mixed receptor and source model	NO_3^- , SO_4^{2-} , NH_4^+ , Cl^- , organic aerosol (OA), Si, K, Ca, Cr, Mn, Fe, Ni, Cu, Zu, As, Se, Ba, Pb	(ordered by significance) residential combustion, traffic emissions, and industry emissions

Table 2. Cont.

Reference City Country	Model	Source Apportionment	Emission Sources
[25] Northwest coast Spain	source model	Al, As, Ba, Co, Cr, Cu, Fe, Mn, Ni, Pb, Sr, V, Zn, Cl^- , NO_3^- , SO_4^{2-} , NH_4^+ , Na^+ , K^+ , Ca^{2+} , Mg^{2+} , $\text{C}_2\text{O}_4^{2-}$	marine sources, crustal sources, and anthropogenic sources (including road dust)
[26] Downtown Bangkok Thailand	source model	$\text{PM}_{2.5}$ data and traffic volume data of different types of vehicles	(traffic sources) expressway (32%), mainly contributed by heavy-duty vehicles; and urban roads (17.5%), evenly contributed by light-duty and heavy-duty vehicles
[27] Montréal Canada	source model	EC, OC, Al, Fe, K, Mg, Na, Zn, Ni, Sr, Cu, Ti, Co, Cr, V, Mn, Cd, Mo, Sb, Pb	SIA and SOA (39%); traffic, road dust, industrial, and cooking emissions (23%)
[28] Guangdong China	tracer model	SO_4^{2-} , NH_4^+ , NO_3^- , OC, EC	industry sources (22.5%), mobile sources (19.3%), agriculture sources (17.5%), biogenic (15.3%), residential (13.1%), dust (6.0%), power plant (4.1%), and others (2.2%)
[29] Guangzhou China	tracer model	radiocarbon and SOA tracers	fossil carbon contributing to EC (71%), WIOC (40%), and WSOC (33%)
[30] PRD China	hybrid of all three models	EC, OC, Na^+ , Cl^- , SO_4^{2-} , NH_4^+ , NO_3^- , Al, Fe, Ti, Cu, Sb, Cd, Co, and species tracers	Fe contributors: area sources (41%), power generation (26%), mobile emissions (26%); Cu contributors: area sources (31%), industry sector (27%), power generation (20%); Mn contributors: area sources (40%), outside PRD (23%), power generation (17%)

4. Pollution Mass Prediction

$\text{PM}_{2.5}$ mass prediction is a widely studied application for a better civic service of air quality awareness so as to mitigate the risk of exposure to high $\text{PM}_{2.5}$ emissions. The research focus could be on outdoor or indoor $\text{PM}_{2.5}$ mass prediction, and some researchers were interested in analyzing the infiltration rate between outdoor and indoor $\text{PM}_{2.5}$ mass. From the review of the relevant articles, most existing works predicted outdoor $\text{PM}_{2.5}$ mass, and the adopted approaches fell into four types of methods, namely, the source dispersion method, the statistical method, machine learning, and hybrids of multiple approaches. The articles using these approaches were reviewed as follows.

If the source apportionment of the pollution was available, the mass of particular chemical species could be predicted by resolving the chemical molecular balance and the backward trajectory analysis from the indicated sources. A source dispersion method was proposed in Dhyni et al. [31] for forecasting $\text{PM}_{2.5}$ mass along urban highways in Delhi, India. In addition to the measured molecular mass and settling velocity, complex factors were considered in the analysis, such as road types; various vehicle models, sizes, and ages; motorized or non-motorized vehicles; wind speeds and directions; relative humidity; and temperature. These factors were incorporated into the source dispersion analysis. The result revealed the $\text{PM}_{2.5}$ mass contribution from various vehicle models, sizes, and ages by tracing the chemical markers. Di et al. [32] apportioned the chemical components of $\text{PM}_{2.5}$, including sulfate, nitrate, OC, EC, ammonium, sea salt, and dust. The authors incorporated 11 meteorological variables and six land use terms for the imputation of missing $\text{PM}_{2.5}$ data in a northern USA area during 2001–2010. The adopted meteorological variables were

temperature, precipitation, atmospheric pressure, relative humidity, visibility, and wind speed, while the land use terms included the percentage of urban areas, population density, road density, elevation, and distance to the nearest monitoring site, which were useful for predicting local emissions.

If the provision of source apportionment was not available, PM_{2.5} concentrations could be predicted via statistical analysis or machine learning approaches. For statistical approaches, Yin et al. [33] estimated daily PM_{2.5} concentration from the existing data of PM₁₀ and the meteorological conditions in Beijing. The relationship between daily PM_{2.5}, PM₁₀, and mean relative humidity was finally identified by a multiple nonlinear regression (MNLR) model. Moisan et al. [34] proposed a PM_{2.5} concentration forecasting method based on dynamic multiple linear equations. Carbon monoxide and four meteorological variables were included in the model and 24 equations for each hourly interval in a day were derived. The prediction accuracy with data from Santiago, Chile, was competitive with several nonlinear forecasting models, such as artificial neural network (ANN) approaches. Time series analysis without external factors is also a viable approach for PM_{2.5} forecasting. Two popular prediction approaches based on time series analysis are the empirical mode decomposition (EMD) ensemble and the autoregressive integrated moving average (ARIMA) model. The EMD technique decomposes the time series into separate modes for predicting individual trends contained in the modes and combines the predictions with an ensemble. Sun and Li [35] proposed an EMD ensemble for predicting hourly PM_{2.5} in severe haze episodes in Baoding, China. The complex non-stationary time series was decomposed into several modes with different frequencies to reduce the noise. A prediction ensemble was developed to recombine cleaned individual modes for enhancing the forecasting accuracy. On the other hand, an ARIMA model was adopted in Ni et al. [36] to discover the temporal trends contained in the PM_{2.5} time series and to estimate the short-term daily PM_{2.5} concentration in Beijing. In addition to the short-term PM_{2.5} prediction, the authors conducted the correlation analysis with meteorological (rainfall, temperature, relative humidity, and wind speed), pollutant (CO, NO₂, SO₂, and PM₁₀), and social media factors (microblog data) based on the multivariate statistical analysis method. They disclosed that the most influential factors contributing to air pollution in Beijing were wind speed and concentrations of CO, NO₂, and PM₁₀, and disclosed the daily number of microblog entries with the keyword 'Beijing'. Zhao et al. [37] proposed an improved ARIMA method integrating Akaike information criterion and an improved grid search for forecasting PM_{2.5} mass in Beijing. The result showed that the improved ARIMA significantly outperformed the standard ARIMA on widely used measures for prediction performance.

Machine learning is a promising prediction approach which has manifested a high forecasting accuracy in many applications. Among many application domains, PM_{2.5} mass forecasting is a leading outlet for applying machine learning prediction approaches. Mogollón-Sotelo et al. [38] trained a support vector machine (SVM) on days with high PM_{2.5} concentrations and those with fast concentration changes to forecast PM_{2.5} concentrations in Bogotá. An ensemble model was established in Di et al. [39] to combine PM_{2.5} estimates from ANN, random forest, and gradient boosting. After training with predictor variable data from 2000 to 2015, good prediction performance was obtained at every 1 km × 1 km grid cell in the contiguous United States. The long short-term memory (LSTM) model was employed in Xiao et al. [40] for predicting the daily PM_{2.5} concentration in the Beijing–Tianjin–Hebei (BTH) region, China. The spatiotemporal information of the surrounding monitoring stations, including their measured concentrations, latitude and longitude, the distance to the central site, and meteorological conditions (temperature, wind speed, wind direction, mean sea level pressure, dew point temperature, and total

column water vapor), is fed into an MLP to adjust the weighting of the central site PM_{2.5} time series, which is then analyzed by the LSTM to address the spatiotemporal dependency. A convolutional neural network (CNN) and an LSTM were deployed in Qin et al. [41] for forecasting of the next-day 24 hourly PM_{2.5} concentrations in 14 sites in Shanghai. The CNN is designed to learn the spatial relationships of meteorological variables and the LSTM is used to model the temporal correlation between pollutants. An empirical study of several machine learning approaches for the prediction of PM_{2.5} in Sanandaj, Iran, was conducted by Ausati and Amanollahi [42]. Four different methods, namely, the ensemble empirical mode decomposition and general regression neural network (EEMD-GRNN), adaptive neuro-fuzzy inference system (ANFIS), principal component regression (PCR), and multiple linear regression (MLR), were evaluated. Among them, the EEMD-GRNN model gave the best prediction accuracy.

To boost the prediction accuracy, an increasing number of hybrid methods were recently developed. For instance, in a case study of Mexico City [43], an ant colony optimization (ACO) algorithm was applied to combine three ANFIS-based methods to enhance the performance of PM_{2.5} prediction. The prediction performance obtained by each individual model was used as an incentive to seduce the ants to follow the trail. As learning converged, the ACO was able to find the near-optimal strategy which selected the best predictor to apply for the prediction. A time series generalized autoregressive conditional heteroscedasticity (GARCH) model was integrated with an SVM in Wang et al. [44] to capture linear and nonlinear panel information for the forecasting of PM_{2.5} concentrations in Shenzhen, China. Zhang et al. [45] proposed a multi-model fusion method for predicting PM_{2.5} in Beijing with monitored data during the period between 1 January and 31 October 2018. Multiple regression models were adopted, and the prediction results were sent to a backpropagation neural network (BPNN) for final prediction fusion. The result showed that the multi-model fusion method surpassed a single model. A new hybrid PM_{2.5} forecasting method combining the variational mode decomposition method (VMD), ARIMA, and a CNN was proposed in Cai et al. [46]. The experimental results with the benchmark dataset of Lanzhou, China, showed that the developed hybrid was superior to individual embedded models.

Table 3 summarizes the applied methodologies, independent variables, and predicting time resolution of the reviewed articles.

Table 3. Summary of reviewed articles for outdoor PM_{2.5} mass prediction.

Reference City Country	Methodology and Applied Technique	Independent Variables	Predicting Time Resolution
[31] Delhi India	Source dispersion CALINE4 simulation	road types; vehicle information; and meteorological variables	Hourly PM _{2.5} within a day
[32] Northeastern USA	Source dispersion Chemical transport simulation	11 meteorological variables; and six land use variables	Imputation of daily PM _{2.5} , EC, OC, nitrate, sulfate, ammonium, dust, and sea salt from 2001 to 2010
[33] Beijing China	Statistical method Multiple nonlinear regression	PM ₁₀ and mean relative humidity	Daily PM _{2.5} within six months
[34] Santiago Chile	Statistical method Dynamic multiple linear equations	carbon monoxide, temperature, relative humidity, wind direction, and wind speed	Hourly PM _{2.5} in five months
[35] Baoding China	Statistical method EMD ensemble	time series forecasting without external variables	Hourly PM _{2.5} in severe haze episodes

Table 3. Cont.

Reference City Country	Methodology and Applied Technique	Independent Variables	Predicting Time Resolution
[36] Beijing China	Statistical method ARIMA	time series forecasting and correlation with meteorological, pollutant, and social media factors	Daily PM _{2.5} within a month
[37] Beijing China	Statistical method Improved ARIMA	time series forecasting without external variables	Daily PM _{2.5} within nearly three years
Mogollón-Sotelo [38] Bogotá Colombia	Machine learning SVM	time series forecasting without external variables	PM _{2.5} for days with high and rapidly changing concentrations
[39] 1 km × 1 km grids in contiguous USA	Machine learning Ensemble of ANN, random forest, and gradient boosting	satellite data, elevation, chemical transport model predictions, and meteorological and land use variables	Daily PM _{2.5} from 2000 to 2015
[40] BTH China	Machine learning MLP and LSTM	distance to neighboring sites and their PM _{2.5} , latitude and longitude, and weather conditions temperature, relative humidity, wind direction, wind speed, precipitation, CO, CO ₂ , NO, NO ₂ , SO ₂ , and PM ₁₀	Daily PM _{2.5} within 2017
[41] Shanghai China	Machine learning CNN and LSTM	temperature, relative humidity, wind speed, precipitation, atmospheric pressure, CO, O ₃ , NO ₂ , SO ₂ , and PM ₁₀	Hourly PM _{2.5} within a day
[42] Sanandaj Iran	Machine learning Comparisons of EEMD-GRNN, ANFIS, PCR, and MLR	time series forecasting without external variables	Daily PM _{2.5} within 30 days
[43] Mexico City Mexico	Hybrid of three ANFIS-based methods	SO ₂ , NO ₂ , CO, O ₃ , dew point, relative humidity, wind speed, barometric pressure, visibility, temperature, and previous hour PM _{2.5}	Hourly PM _{2.5} within a day
[44] Shenzhen China	Hybrid of GARCH, ARIMA, and SVM	meteorological variables	Next hour PM _{2.5} within a day
[45] Beijing China	Hybrid of multiple regression models and BPNN	time series forecasting without external variables	Hourly PM _{2.5} within a day
[46] Lanzhou China	Hybrid of VMD, ARIMA, and CNN		Hourly PM _{2.5} within a day

In addition to the PM_{2.5} mass prediction in the outdoor environment, some researchers have contemplated the indoor PM_{2.5} analysis as well. People usually spend much more time in indoor places, such as offices and homes, than outdoor spaces. The research of indoor PM_{2.5} analysis is focused on the relationship between indoor and outdoor concentrations and the prediction of indoor PM_{2.5} mass, such that appropriate air ventilation control can be automated. Lee et al. [47] analyzed the relationship between the indoor PM_{2.5} infiltrated from outdoor PM_{2.5} in the greater Boston area. They used sulfur as a unique tracer for indoor PM_{2.5} of outdoor origin because sulfur seldom has indoor sources. Among a set of meteorological variables, temperature was identified as the most significant factor by applying linear regression models. It was found that the increases in temperature were highly associated with an increased indoor/outdoor PM_{2.5} infiltration rate. Li et al. [48] developed a backpropagation neural network (BPNN) model combined with an adaptive multi-objective particle swarm optimizer (AMOPSO) to predict and control indoor CO₂

and PM_{2.5} concentrations. The indoor airflow data of CO₂ and PM_{2.5} concentrations were characterized by the computational fluid dynamics model and fed forward to the BPNN for prediction. If the predicted concentrations exceeded the tolerance level, the AMOPSO was executed to automatically adjust the heating, ventilation, and air conditioning (HVAC) systems to improve the indoor air quality. The result showed that the proposed method could reduce CO₂ and PM_{2.5} concentrations by 30.5% and 77.1%, respectively. An indoor PM_{2.5} prediction model for a hospital outpatient hall was proposed by Zhou and Yang [49]. To enhance the prediction accuracy, the occupancy level of outpatients was obtained by using image recognition, and both the subjective and objective information were used for weighting. The result showed that the occupancy information could improve the prediction accuracy by 89% compared to the traditional method without using the occupancy information. Li et al. [50] developed a random forest approach for predicting residential indoor PM_{2.5} concentrations in Hong Kong. The information about the households and ventilation settings in 116 households were collected and the outdoor PM_{2.5} concentration was estimated using a land use regression model. The result showed that the most dominant factor for residential indoor PM_{2.5} prediction was the outdoor PM_{2.5} concentration, followed by the household marked number, outdoor temperature, outdoor relative humidity, average household area, and air conditioning. Elbayoumi et al. [51] compared the multivariate linear regression with the feedforward backpropagation (FFBP) model to analyze the seasonal influence on the concentrations of indoor PM_{2.5–10} and PM_{2.5} in twelve naturally ventilated schools in the Gaza Strip, Palestine. The comparison showed that the FFBP outperformed the multivariate linear regression, especially for predicting the indoor PM_{2.5} concentration in fall.

5. Association Analysis

One of the important problems for air pollution analysis is the disclosure of the associations between pollution and the subjects of interest. Researchers have disclosed natural and anthropogenic factors which exacerbate pollution concentrations. The influential natural factors range from meteorological variables, climate change, volcanic emissions, to wildfires, while the influential anthropogenic factors contributing to air pollution include population activity density, holidays, landmarks (temples, gas stations, markets, etc.), and land uses (such as agricultural, industrial, business, and residential areas). Conversely, it has been evidenced that air pollution has severe impacts on the natural environment and human health. A high quantity of PM_{2.5} concentrations increases the fatality of living things, deteriorates the air visibility, and decreases solar irradiation. By disclosing the associations between the air pollution and these factors, one can predict the air quality and alleviate the impacts from air pollution. In the following, this review will comment on the articles which explored the pollution observational data and applied statistical analysis or machine learning approaches to find the underlying associations between air pollution and various natural and anthropogenic factors.

The natural factors contributing to air pollution have been identified in the literature. One of the most influential natural factors is meteorological variables. However, the significance of the influence of these factors varies with climate and spatial and temporal dimensions. Therefore, the indicated associations should be carefully explained with the underlying case study. For instance, Wang et al. [52] investigated the 14 most polluted cities in China during January 2017 and claimed that the height of the atmospheric boundary layer, wind speed, and precipitation most influenced the PM_{2.5} concentration, while temperature had impacts on PM_{2.5} concentration but only under certain conditions. A case study in Yunlin County, Taiwan, was conducted in Hsu and Cheng [53], which stated that the meteorological variables, namely, the hourly precipitation, temperature, wind direction,

and wind speed had significant influences on $PM_{2.5}$ concentrations. By applying clustering algorithms with the meteorological variables, six weather patterns were identified. The $PM_{2.5}$ concentrations observed on the days of each weather pattern were analyzed. The results showed that high $PM_{2.5}$ concentrations were associated with the weather patterns of a continental anticyclone or a stagnant local circulation, and low $PM_{2.5}$ concentrations coexisted with a strong southwesterly monsoonal flow pattern. A case study for the prediction of $PM_{2.5}$ in Sanandaj, Iran, was conducted in [42]. Meteorological variables including minimum temperature, maximum temperature, atmospheric pressure, precipitation, wind speed, and relative humidity were tested by four machine learning models. It was identified that the maximum temperature and the wind speed were the most influential variables responsible for the increase in $PM_{2.5}$ concentrations in Sanandaj, Iran. Yin [54] applied association rule mining to identify the most influential meteorological factors on the most polluted days in Puli township, located in central Taiwan. They disclosed that the most polluted days coexisted with a low temperature and medium relative humidity, with the rule confidence being 0.2485 and the lift being 2.17, measured values which are both very significant. Air pollution has a high correlation with climate change since they share many contributing sources, such as black carbon and methane. Tsai et al. [55] studied the climate change-induced meteorological impacts on $PM_{2.5}$ in Taiwan. Their simulations under the 2 °C and 4 °C global warming scenarios disclosed that climate change worsened the $PM_{2.5}$ concentrations in Taiwan in cold seasons. However, the worsening was mitigated in days with northeasterly winds and nighttime warming. Danek et al. [56] conducted a geostatistical case study of Krakow, Poland, to study the influence of meteorological factors and terrain on air pollution concentration and migration. The data were collected from 100 LCSs during the COVID-19 lockdown period to reduce the background noise. The identified influential factors were wind speed and terrain elevation and the main source contributing to the air pollution in Krakow was the solid fuel heating outside the city. Sometimes terrain is the most critical factor, which is not easy to distinguish using machine learning approaches. The case study in Krakow showed that the Wieliczka Hills at the southern side form a natural terrain barrier resulting in a lower concentration of pollutants associated with the southerly winds. Danek et al. [56] also found that the ratio between $PM_{2.5}$ and PM_{10} was a useful clue to help distinguish if the pollutants were generated from anthropogenic or natural sources.

Natural volcanic emissions and wildfires also cause sudden increases in $PM_{2.5}$ concentrations. Reikard [57] investigated the influence of emissions from Kilauea volcano on the air pollution in Hawaii. The volcanic emissions were used as inputs in several models to predict SO_2 and $PM_{2.5}$. The result showed that simple regressions performed best for short-term $PM_{2.5}$ prediction, while ARIMA predicted more accurately for longer term $PM_{2.5}$ or an arbitrary time horizon for the SO_2 series. Carreras-Sospedra et al. [58] studied the impact of clustered wildfires on air pollution and health in Northern California during the period from 16 August to 28 October of 2020. The wildfire impacts on O_3 and $PM_{2.5}$ were used to quantify the health impacts. It was found that the wildfire activity increased the $PM_{2.5}$ mass by 38 times compared to all days. The result indicated that wildfires increased the daily maximum 8 h average of O_3 by 10 ppb and caused up to 10 times higher $PM_{2.5}$ than the standard. Wildfires caused an additional 466 deaths in this period and accounted for a 1% rise in mortality.

Anthropogenic factors involving various socioeconomic activities can cause different significances of influence on air pollution. For instance, some particular air pollution events are related to holidays because of the existence of significant differences between anthropogenic activities on holidays and workdays. Tan et al. [59] investigated the holiday effect of Chinese New Year (CNY) on the variations of six air pollutants over six divisions

in Taiwan. CNY is the longest national holiday which lasts nearly a week every year. The variation of pollutant mass in the 10 day period before and after CNY, and the days within CNY, was analyzed. They empirically found that the holiday effect was more profound in a division if its urbanization degree was higher. The urbanization degree was measured by the population and vehicle number, which are related to anthropogenic sources. Similarly, another empirical case study of the CNY firework impact on air pollution was conducted in Lanzhou, China [60]. The pollutant concentration in the peak firework period (00:00–00:10) of CNY was approximately two to six times that in the 14 day period before CNY depending on the particle size. Yin [61] proposed an association rule mining approach to identify the association between high $PM_{2.5}$ mass and CNY's eve. It was found that the probability of observing a high $PM_{2.5}$ concentration at CNY's eve was more than four times higher than the mean probability over all days in 2017–2019. Li et al. [62] showed that holidays have cross-country pollution transboundary effects. Using pollution data from 100 and 120 monitoring stations located in China and South Korea, respectively, the result showed the association that the air quality in South Korea was improved by 5.5% during the Chinese holidays and by 1.6% vice versa.

Human-built landmarks also attract anthropogenic activities that promote $PM_{2.5}$ concentrations. Yin [61] conducted association mining between pollution concentrations and the landmarks of interest at Puli, Taiwan. They investigated $PM_{2.5}$ concentrations at the microsites near to the landmarks of interest. The identified influential landmarks according to the decreasing order of the association confidence value are night markets, crossroads, paper mills, temples, and highway exits. This order of influence of significance may be due to the local people's living style at Puli, which is a rural basin town in the mountains. Most citizens are farmers, paper mill operators, or shopkeepers and their daily activities are mostly in town. There are no nightclubs. The night markets are a popular place for the citizens to find food and drink after work. Puli is also one of the towns which has the highest density of temples in Taiwan. This unique living style explains why night markets, crossroads, paper mills, and temples are among the most influential anthropogenic factors for increasing $PM_{2.5}$ concentrations. There is one highway connecting Puli to western metropolitan cities. The highway traffic is severely jammed on weekends because many people from cities crowd into Puli for sightseeing. On weekdays, however, few people commute between cities and Puli. This phenomenon can be attributed to why the contribution of highway exits is less than that of other landmarks. Singh et al. [63] presented a prediction method for population exposure to $PM_{2.5}$ concentrations per unit area and its dependence on anthropogenic activity in London. They considered anthropogenic activity in three microenvironments, namely, residential, work, and transport, and the result showed that 85% of the population exposure to $PM_{2.5}$ concentrations occurred at home and work, and 15% in transport. It also showed that the population exposure in the central London was nearly three times greater than that in suburban regions.

The land use planning also influences the level of local $PM_{2.5}$ concentrations. Various land uses, such as business districts, industrial and residential areas, forests, agricultural, and highlands, potentially incur specific socioeconomic activities which manifest air pollution of different significances. The impact of land use on air pollution was studied in the literature. Di et al. [32] adopted land use terms, including the percentage of urban areas, population density, road density, and elevation, for predicting local emissions in a northern USA area during 2001–2010. The accurate prediction indicating the association between land use terms and local air pollution was significant. Yousefi et al. [64] disclosed long-term $PM_{2.5}$ trends in China. The $PM_{2.5}$ concentration increased during 1980–2007, mainly due to urbanization and deforestation. It changed to decreasing during 2007–2020, which was attributed to air pollution control policies enforced by the government. Zhao et al. [65]

conducted a study investigating the influence of urban land use on PM_{2.5} concentration by using a deep learning simulation approach. The study was conducted with the land use and air pollution data in 2016 for Wuhan, China. The simulation results showed that industrial land contributed the most to pollutants, while urban road transportation exhausts vehicle emissions. Yin [61] investigated the impact of four land use types, namely, the business district, residential area, agricultural fields, and highlands, on air quality. The significance of the association showed that the business district had the greatest impact on increasing air pollution. The residential area had a mild impact on contributing to a high PM_{2.5} concentration. The highland and the agricultural land enjoyed better air quality as compared to the mean background emissions of all land use types.

In addition to the noted influential factors contributing to air pollution, the concentration of PM_{2.5} reversely affects anthropogenic and natural phenomena. There have been several studies revealing the association between air pollution and human diseases. Sadat et al. [1] applied the association rule mining approach to disclose the spatial relationships between asthmatic allergies and air pollution. The case study, located in Tehran, Iran, showed that the living distance to parks and roads may affect the air pollution and asthmatic allergies in December. Milicevic et al. [66] presented an interesting study disclosing a relationship between COVID-19 transmissibility and air pollution levels. By using linear regressions, random forest, and gradient boost, the PM_{2.5} level was been shown to be a viable predictor of the basic reproduction number (R_0) in USA states. The result also showed that PM_{2.5} level variations were estimated to change R_0 by about 30%. Khatibi and Karampour [67] proposed a stacked ensemble of CNN and LSTM to predict the number of patient admissions due to psychological disorders by using air pollutants and weather condition descriptors as input variables. The data were collected from 21 March 2014 to 18 January 2017 in Iran. The experimental results showed that good prediction can be obtained with outstanding levels of RMSE, MAE, and R^2 . It was reported in Wan et al. [68] that air pollution can have an influence on successful pregnancy with frozen embryo transfer (FET). The air pollution data and the women's FET data between 2015 and 2018 in Shanghai were added to a random forest model. The result indicated that for one-month of exposure before FET, NO₂ was negatively associated with intrauterine pregnancy.

The literature also showed the influences of air pollution on environmental sources. Fan et al. [69] chose the six air pollutants PM_{2.5}, PM₁₀, SO₂, NO₂, CO, and O₃, and the air quality index (AQI), and fed them into an SVM model for analyzing their individual and combined effects on the prediction of global solar radiation (R_s) and diffuse solar radiation (R_d). The result showed that individually, AQI was the most powerful predictor for both R_s and R_d prediction. For a combined predictor with two pollutants, either PM₁₀ or PM_{2.5} combined with O₃ performed the best for R_s and R_d prediction. Vakili et al. [70] deployed a Multi-layer Perceptron (MLP) neural network using temperature, relative humidity, wind speed, PM_{2.5}, and PM₁₀ as inputs to predict the R_s and R_d of Tehran, Iran, on the sunny days in the years 2012 and 2013. Song et al. [71] quantified the effect of PM_{2.5} in decreasing the power generation from available solar irradiation in Hong Kong. Their analysis showed that the global irradiance was decreased by more than 5% when the PM_{2.5} concentration exceeded 33.5 µg/m³, resulting in 7.00 and 9.73% losses in the energy produced by crystalline silicon and thin-film PV, respectively.

Table 4 summarizes the features of the reviewed articles, including their applied methodologies, identified factors, and whether these factors are influencing or influenced, and natural or anthropogenic.

Table 4. Summary of reviewed articles for association analysis.

Reference City Country	Methodology and Applied Technique	Influencing/Influenced Natural/Anthropogenic	Identified Factors
[52] 14 cities China	Community Multi-scale Air Quality (CMAQ) model	Influencing Natural	atmospheric layer, wind speed, and precipitation
[53] Yunlin County Taiwan	Hierarchical clustering refined by K-means	Influencing Natural	hourly precipitation, temperature, wind direction, and wind speed
[44] Sanandaj Iran	Ensemble empirical mode decomposition and general regression neural network	Influencing Natural	maximum temperature and wind speed
[54] Puli Taiwan	Association rule mining	Influencing Natural	temperature and relative humidity
[56] Krakow Poland	Geostatistical approach	Influencing Natural	wind speed and terrain elevation
[57] Hawaii USA	Regressions ARIMA	Influencing Natural	Kilauea volcano emissions
[58] Northern California USA	Community Multi-scale Air Quality (CMAQ) model	Influencing Natural	clustered wildfires
[59] Six divisions Taiwan	Statistical analysis on empirical data	Influencing Anthropogenic	Chinese New Year
[60] Lanzhou China	Statistical analysis on empirical data	Influencing Anthropogenic	fireworks
[61] Puli Taiwan	Association rule mining	Influencing Anthropogenic	Chinese New Year
[62] China and South Korea	Statistical analysis on empirical data	Influencing Anthropogenic	cross-country holiday effect
[61] Puli Taiwan	Association rule mining	Influencing Anthropogenic	night markets, crossroads, paper mills, temples, and highway exits
[63] London UK	Multiple-source Gaussian dispersion approach	Influencing Anthropogenic	residential, work, and transport microenvironments
[32] Northern USA	Chemical transport model and land use regression Modern-Era	Influencing Anthropogenic	percentage of urban areas, population density, road density, and elevation
[64] Six regions China	Retrospective Analysis for Research and Applications, version 2 (MERRA-2)	Influencing Anthropogenic	urbanization and deforestation
[65] Wuhan China	Deep learning	Influencing Anthropogenic	industrial land use and road transportation
[61] Puli Taiwan	Association rule mining	Influencing Anthropogenic	business district, residential area, agricultural fields, and highlands
[1] Tehran Iran	Association rule mining	Influenced Anthropogenic	asthmatic allergies
[66] Contiguous USA	Linear regressions, random forest, and gradient boost	Influenced Anthropogenic	COVID-19 transmissibility
[67] Tehran Iran	CNN and LSTM	Influenced Anthropogenic	psychological disorders
[68] Shanghai China	Random forest	Influenced Anthropogenic	pregnancy with frozen embryo transfer

Table 4. Cont.

Reference City Country	Methodology and Applied Technique	Influencing/Influenced Natural/Anthropogenic	Identified Factors
[69] Beijing China	SVM	Influenced Natural	global and diffuse solar radiation
[70] Tehran Iran	Multi-layer Perceptron	Influenced Natural	global and diffuse solar radiation
[71] Hong Kong China	Statistical analysis on empirical data	Influenced Natural	global solar irradiation

6. Opportunities and Challenges

From the review of the selected articles published in the last decade, it was identified that some special topics still have large room for exploration. These topics, including air pollution spatiotemporal analysis, maintenance of monitoring stations, pollution control policies, and ideologies, grow increasingly important at the next stage of research for air pollution analysis. In the following, this review highlights the opportunities and challenges of these promising topics.

6.1. Spatiotemporal Analysis

Spatiotemporal analysis for air pollution refers to the research on realizing the variation in pollution mass or its relationship to other factors as time and location change. From this review, it was observed that some features in the existing works revealed challenges and opportunities for future works. (1) Attributed to the high complexity of considering simultaneously both spatial and temporal dimensions, most existing works for spatiotemporal analysis handled the two dimensions separately. (2) To include a large geographical coverage of PM_{2.5} concentrations, some works conducted spatiotemporal analysis by using the AOD satellite images. (3) Multi-scale projection is a viable technique to explore the variations in different resolutions for spatiotemporal analysis. The multi-scale metric can be applied on spatial, temporal, or pollution levels.

For instance, Munir et al. [72] analyzed separately the spatial and temporal PM_{2.5} trends in Saudi Arabia. The ground surface PM_{2.5} concentrations were estimated by the AOD data and the result showed that the PM_{2.5} concentrations varied spatially with a negative trend from south to north, and a positive trend in PM_{2.5} levels was observed temporally for most cities. Bai et al. [73] conducted a separate spatial and temporal trend analysis for PM_{2.5} concentrations in China using the satellite-based and ground-measured PM_{2.5} data. The result indicated that the PM_{2.5} concentrations increased before 2008 and have decreased since then. The most significant decrease was observed in central China, especially in wintertime.

Bai et al. [74] applied the multi-scale technique on spatial and temporal dimensions separately. Overall, 2013 was the turning point year for PM_{2.5} trends in China from the period of 2000–2020. The PM_{2.5} variations were higher at the city- and county-levels than the province-level, and they were also higher in eastern and western China than the other regions in China. Yue et al. [75] employed the multi-scale technique on the spatial landscape dimension to analyze the dust and sea salt-removed pollution dynamics on the globe. It was found that the global pollution area increased by a factor of 2.43 from 1,146,800 to 3,929,800 km² in the period of 2000–2016, and the most densely distributed pollution was observed in India and China. It also revealed that the pollution area became more geographically fragmented in the same period. Saez et al. [76] assumed independence between spatial and temporal dimensions and applied a hierarchical Bayesian spatiotemporal model

to predict the long-term and short-term trends of PM_{2.5} distributions in Catalonia, Spain. Temporally, the long-term daily O₃ increased by 3.97% in 2019 compared to 2011–2018, while the other investigated pollutants decreased. For short-term exposure, daily NO₂ and PM₁₀ increased from 9 September, while O₃ and PM_{2.5} decreased after 9 September. As for spatial analysis, the predictive performance started to deteriorate if observations from less than 40 monitoring stations were used for inference.

Zareba et al. [77] applied K-means algorithm with Dynamic Time Warping (DTW) for enhancing spatiotemporal air pollution pattern analysis. The studied field was in Krakow, Poland, with the one-year data of hourly PM₁₀ collected from almost 100 sensors. The clustering result showed that there existed great differences between the average and maximum values of PM₁₀ concentrations. Moreover, the smoother clusters after kriging data significantly facilitated the interpretation. Rincon et al. [78] conducted a statistical spatiotemporal analysis of PM₁ and PM_{2.5} concentrations and their key influencing factors at Guayaquil city, Ecuador. By applying bivariate analysis, and multivariate linear and logistic regression, it was shown that PM₁ concentrations were decreased in rain and the flat orography of the cement plant sector, and they were increased in traffic jams and vegetation fires. While PM_{2.5} concentrations were increased in the dry season and the industrial sector, they were decreased by the cement plant. Di et al. [79] established a convolutional neural network (CNN) model which aggregated neighboring information into the CNN to account for spatial and temporal autocorrelation. The CNN was trained for the continental United States from 2000 to 2012. The result showed the model had a high R² value with monitoring data in various tests, including short-term, long-term, geographical divisions, elevation variations, and low PM_{2.5} concentrations.

In addition to the previously noted works, some novel approaches were proposed recently, and they gave promising results for genuine spatiotemporal analysis simultaneously working on both spatial and temporal dimensions. For example, association rule mining can be used as a technique to conduct spatiotemporal analysis if the time and location information are both included in the mining dataset. A multi-scale spatiotemporal association rule mining approach was proposed in Yin [61] to disclose the relationships between PM_{2.5} concentrations and anthropogenic festivals and landmarks. Multiple scales on spatial-, temporal-, and PM_{2.5}-level dimensions were experimented with to identify the most influential factors which result in different PM_{2.5} levels. Another promising approach for air pollution spatiotemporal analysis was proposed in Yin [54], where a multimedia-based spatiotemporal analysis framework was developed for extracting and clustering air pollution episodes. The spatiotemporal series of PM_{2.5} concentrations were considered as a video stream and the shot transition detection technique could be applied to detect the cut between various air pollution episodes, such as cumulating, scattering, dispersion, and stagnation. Each extracted spatiotemporal episode was represented by a gait energy image (GEI) which had high/low gray levels at constantly serious/no pollution regions and median gray levels at the regions where pollution transported through. To realize how often and when the episode appeared, the GEIs were grouped into clusters and each cluster contained quasi-homogeneous pollution episodes which emerged at different times. The multimedia-based spatiotemporal analysis framework is very useful to perceive the scenarios which result in particular pollution patterns. Yin further extended this framework in Yin [80] to add two new elements: the spatiotemporal retrieval and feature analysis of air pollution episodes. The spatiotemporal retrieval of air pollution episodes offers the user with a query-by-sketch mechanism which allows the user to sketch the spatial relationships between land objects (e.g., rivers, mountains, temples, gas stations, etc.) and pollution objects (regions with a certain degree of pollution) and retrieve the most similar pollution episodes stored in the database. The user can further click on a retrieved episode

to see the video shot of the contiguous concentration images. Various spatiotemporal 3D convolutional filters for feature extraction are provided to automatically detect the dynamic 3D features of the pollution episode.

6.2. Maintenance of Monitoring Stations

Robust analysis of air pollution relies on contiguous data from monitoring stations. As previously noted, supersites are costly and mostly deployed at seriously polluted places. To increase the density of the monitoring stations, low-cost microsites are implemented by participatory parties and interconnected via an Internet of Things (IoT). The microsite IoT promotes the civic science of air pollution and increases the public awareness of pollution reduction actions. However, the maintenance for a huge number of microsites to lengthen the monitoring uptime is a very challenging job. It becomes laborious and costly to perform maintenance of the IoT devices considering various maintenance resource constraints and IoT reliability requirements. In traditional manufacturing, equipment maintenance intends to increase the manufacturing uptime and reduce the delay due to equipment failures. There are a variety of maintenance tasks, some of which are low-cost and easily implemented, such as inspections, cleaning, adjusting, and lubricating, while other tasks may be expensive and complicated, ranging from repairs, replenishment, upgrades, fault diagnosis, to entire replacement. Maintenance planning focuses on developing strategical policies and tactical methodologies for efficient and economical management of equipment maintenance. There are two main policies for making maintenance plans [81,82], namely, the corrective maintenance (CM) and preventive maintenance (PM) policies. CM performs maintenance activities only when equipment halts operations due to failures, so CM is passive and responsive to unplanned events which can incur great costs and lengthen the system breakdown if no anticipated maintenance plan was made to prevent or predict such equipment faults. On the other hand, PM makes an anticipated plan based on the estimated mean time between failures (MTBF) of equipment such that appropriate maintenance tasks can be performed in advance, so PM is proactive and programmable to avoid expected failures before they take place.

By contrast to maintenance in traditional manufacturing, the maintenance plan for a wide-area outdoor air quality monitoring IoT has additional challenges including, but not limited to, the following: (1) In traditional manufacturing, the amount of equipment is limited and equipment is located within a factory and easy to access, while the wide-area outdoor air quality monitoring IoT contains thousands of microsites and spans a large geographical area where the distance between maintenance sites is measured in kilometer scale. A viable maintenance plan should consider deploying a fleet of engineering vehicles to carry sufficient engineers and maintenance materials for operations. It is perplexing to derive the most economical maintenance plan to select the microsites for maintenance in order to guarantee the IoT service quality. The visiting sequence of the chosen maintenance sites also becomes very critical to shorten the vehicle traveling distance in order to save the working hours and consumed fuel. The two noted tasks, the subset selection problem and the vehicle routing problem, are both NP-hard, implying the necessity for designing appropriate heuristics. (2) As the sensors of an air quality monitoring IoT work outdoors, the failure rate may vary depending on the local environment and weather conditions at the site. More information needs to be tallied during maintenance activities to re-estimate the sensor's reliability and to adaptively revise the maintenance plan. (3) In traditional manufacturing, the equipment operates in a serial or parallel fashion. The system's overall reliability is estimated based on the reliability of individual pieces of equipment and their cooperating architecture. However, for an air quality monitoring IoT, the sensor failure will not immediately cause system breakdown but will deteriorate the IoT service level. A new

objective measurement for the maintenance of an air quality monitoring IoT needs to be defined such that the criticality of a sensor failure can be appropriately gauged.

6.3. Policies and Ideologies

Due to the tremendous economic losses and health risks caused by air pollution, an increasing number of state authorities have made control policies to reduce hazardous emissions for a robust and sustainable future. However, compared to other fields of air pollution research, there is less research addressing the effectiveness of the control policies on emission reduction. A recent review of the impacts of control policies on air pollution was presented in Feng et al. [83], with the relevant articles published between 1998 and 2022. The findings disclosed that China was the major contributor in this research direction. There is still a gap regarding future endeavors such as the synergistic effects of air policies and the coupling of multiple models for policy evaluation. Kumar et al. [84] conducted an evaluation of various control strategies for industrial air pollution sources in the industrial region of Chembur in Mumbai, India. Several control policies, like increment of stack height and the use of low-emission fuel, were proposed to improve the air quality. The study evaluated various scenarios for three settings of the stack heights and two types of low-emission fuel. The result indicated that raising up the stack height by 50% had the biggest reduction in ground level concentration for all pollutants, but caused longer residence times of pollutants in the atmosphere. A preferred policy is using a lower emission fuel which actually reduces the source emissions. Zareba et al. [85] studied the impacts of energy transformation and environmental policy through a multidimensional historical analysis of air pollution. Multiple LCSs were deployed to establish an IoT for monitoring the PM_x concentrations at various locations in Krakow, Poland. Different machine learning approaches were applied to conduct a deeper analysis of the historical data and make efficient predictions of future events. Krakow is one of the cities practicing the strictest air control policies to prohibit the use of solid fuels and establish clean transport zones. The result showed that population growth in Krakow did not substantially increase PM_x concentrations, and energy transition from coal-based to renewable sources was the primary factor in reducing PM_x concentrations.

Another challenging research direction is the analysis of human ideologies for pollution reduction. It is worth studying whether human morality about environmental protection and sustainable society triggers actual actions or behaviors towards helping reduce the emissions in people's daily lives. Very little research on this topic has been observed in the past decades. Shi et al. [86] investigated the determinants that affect household $PM_{2.5}$ reduction behavioral intention in Chinese urban areas. Applying planned behavior theory and norm activation theory to 418 urban households who lived in heavy haze areas, the analysis result revealed that household attitude towards $PM_{2.5}$ reduction, perceived behavioral control, and descriptive norm all affected consequent intention, while subjective norm did not. A following article by Shi et al. [87] conducted an empirical study in China to predict the impacts of psychological factors and policy factors on individual's $PM_{2.5}$ reduction behavior. The empirical study employed the questionnaire survey data of 1007 residents living in severely polluted cities. The result indicated that psychological factors, including an individual's subjective norm, attitude, and self-efficacy, are factors affecting an individual's $PM_{2.5}$ reduction intention, and the policy factors of controllability, economic incentive, and education guidance can facilitate the transformation from intention to actual behavior towards $PM_{2.5}$ reduction.

7. Conclusions

Air pollution has long been one of the most important topics in previous UN global meetings on climate change. The research of air pollution analysis coincides with several goals in the SDGs, namely, good health and well-being (Goal 3); affordable and clean energy (Goal 7); decent work and economic growth (Goal 8); industry, innovation, and infrastructure (Goal 9); sustainable cities and communities (Goal 11); responsible consumption and production (Goal 12); and climate action (Goal 13). This review has the following implications:

- The importance of air pollution analysis has led to an overwhelming number of research articles. At least nine review papers on special topics have been published in the last decade, indicating the growing research interest in the community.
- Through the perusal of existing review papers, it was found that the discussions on some topics such as source apportionment were outdated, and some topics, like mass prediction and association analysis, were overlooked. This review thus aimed at filling the gap between the existing reviews and the state-of-the-art technologies, and the research fields which have not been addressed in the previous reviews but will remain increasingly important in the future.
- This review chose research on three special topics, namely, source apportionment, PM_{2.5} mass prediction, and factor association analysis, published between 2013 and 2024. The reviewed papers were compared according to the characteristics of the topic, such as the applied models or methodologies, the compositional elements of the pollution, the emission sources contributing to the pollution, the studied areas and the time period, the independent variables for prediction, the predicting time resolution, and the identified influencing and influenced factors. Through this comparison study, readers can find research trends from multiple aspects and position their research at the cutting edge.
- This review also identified three promising but challenging research fields, namely, the spatiotemporal analysis of air pollution, the maintenance of monitoring stations, pollution control policies, and ideologies, which emerged to be increasingly important for the future. The three research fields pose opportunities to create innovative and crossover approaches from different disciplines, such as multimedia processing, device operation and maintenance, and psychological and behavioral cognition.

Funding: This research was funded by the National Science and Technology Council of the ROC, grant number NSTC 112-2410-H-130-012.

Conflicts of Interest: The author declares no conflicts of interest.

References

1. Sadat, Y.K.; Karimipour, F.; Sadat, A.K. Investigating the relation between prevalence of asthmatic allergy with the characteristics of the environment using association rule mining. In Proceedings of the 1st ISPRS International Conference on Geospatial Information Research, Tehran, Iran, 15–17 November 2014; pp. 169–174.
2. Mo, L.; Ma, Z.; Xu, Y.; Sun, F.; Lun, X.; Liu, X.; Chen, J.; Yu, X. Assessing the Capacity of Plant Species to Accumulate Particulate Matter in Beijing, China. *PLoS ONE* **2015**, *10*, e0140664. [CrossRef] [PubMed]
3. Liu, Y.-J.; Zhang, T.-T.; Liu, Q.-Y.; Zhang, R.-J.; Sun, Z.-Q.; Zhang, M.-G. Seasonal Variation of Physical and Chemical Properties in TSP, PM₁₀ and PM_{2.5} at a Roadside Site in Beijing and Their Influence on Atmospheric Visibility. *Aerosol Air Qual. Res.* **2014**, *14*, 954–969. [CrossRef]
4. IPCC. *Climate Change 2007: The Scientific Basis, Contribution of Working Group I to the Third Assessment Report of the Intergovernmental Panel on Climate Change*; Houghton, J.T., Ding, Y., Griggs, D.J., Noguer, M., Vander Linden, P.J., Dai, X., Maskell, K., Johnson, C.A., Eds.; Cambridge University: New York, NY, USA, 2007.
5. WHO. Ambient (Outdoor) Air Pollution. 2022. Available online: [https://www.who.int/en/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/en/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health) (accessed on 20 August 2024).

6. Singh, N.; Murari, V.; Kumar, M.; Barman, S.; Banerjee, T. Fine particulates over South Asia: Review and meta-analysis of PM_{2.5} source apportionment through receptor model. *Environ. Pollut.* **2017**, *223*, 121–136. [[CrossRef](#)] [[PubMed](#)]
7. Wang, Y.; Yang, W.; Han, B.; Zhang, W.; Chen, M.; Bai, Z. Gravimetric analysis for PM_{2.5} mass concentration based on year-round monitoring at an urban site in Beijing. *J. Environ. Sci.* **2016**, *40*, 154–160. [[CrossRef](#)]
8. Shukla, K.; Aggarwal, S.G. A technical overview on beta-attenuation method for the monitoring of particulate matter in ambient air. *Aerosol Air Qual. Res.* **2022**, *22*, 220195. [[CrossRef](#)]
9. Day, R.F.; Yin, P.Y.; Huang, Y.C.; Wang, C.Y.; Tsai, C.C.; Yu, C.H. Concentration-temporal multilevel calibration of PM_{2.5} low-cost sensor. *Sustainability* **2022**, *14*, 10015. [[CrossRef](#)]
10. Lewis, A.; Peltier, W.R.; von Schneidmesser, E. *Low-Cost Sensors for the Measurement of Atmospheric Composition: Overview of Topic and Future Applications*; Research Report; World Meteorological Organization (WMO): Geneva, Switzerland, 2018.
11. Hu, X.; Waller, L.A.; Lyapustin, A.; Wang, Y.; Al-Hamdan, M.Z.; Crosson, W.L.; Estes, M.G.; Estes, S.M.; Quattrochi, D.A.; Puttaswamy, S.J.; et al. Estimating ground-level PM_{2.5} concentrations in the Southeastern United States using MAIAC AOD retrievals and a two-stage model. *Remote Sens. Environ.* **2014**, *140*, 220–232. [[CrossRef](#)]
12. Guo, Y.; Tang, Q.; Gong, D.Y.; Zhang, Z. Estimation ground-level PM_{2.5} concentrations in Beijing using a satellite-based geographically and temporally weighted regression model. *Remote Sens. Environ.* **2017**, *198*, 140–149. [[CrossRef](#)]
13. Dehghani, M.; Anushiravani, A.; Hashemi, H.; Shamsedini, N. Survey on air pollution and cardiopulmonary mortality in Shiraz from 2011 to 2012: An analytical-descriptive study. *Int. J. Prev. Med.* **2014**, *5*, 734–740.
14. Mohammed, M.; Song, W.W.; Ma, W.L.; Li, W.L.; Ambuchi, J.J.; Thabit, M.; Li, Y.F. Trends in indoor–outdoor PM_{2.5} research: A systematic review of studies conducted during the last decade (2003–2013). *Atmos. Pollut. Res.* **2015**, *6*, 893–903. [[CrossRef](#)]
15. Dominski, F.H.; Branco, J.H.L.; Buonanno, G.; Stabile, L.; Gameiro da Silva, M.; Andrade, A. Effects of air pollution on health: A mapping review of systematic reviews and meta-analyses. *Environ. Res.* **2021**, *201*, 111487. [[CrossRef](#)] [[PubMed](#)]
16. Li, W.; Lin, G.; Xiao, Z.; Zhang, Y.; Li, B.; Zhou, Y.; Ma, Y.; Chai, E. A review of respirable fine particulate matter (PM_{2.5})-induced brain damage. *Front. Mol. Neurosci. Sect. Brain Dis. Mech.* **2022**, *15*, 967174. [[CrossRef](#)] [[PubMed](#)]
17. Hashemi, F.; Hoepner, L.; Hamidinejad, F.S.; Abbasi, A.; Afrashteh, S.; Hoseini, M. A survey on the correlation between PM_{2.5} concentration and the incidence of suspected and positive cases of COVID-19 referred to medical centers: A case study of Tehran. *Chemosphere* **2022**, *301*, 134650. [[CrossRef](#)] [[PubMed](#)]
18. Krittanawong, C.; Qadeer, Y.K.; Hayes, R.B.; Wang, Z.; Thurston, G.D.; Virani, S.; Lavie, C.J. PM_{2.5} and cardiovascular diseases: State-of-the-Art review. *Int. J. Cardiol. Cardiovasc. Risk Prev.* **2023**, *19*, 200217. [[CrossRef](#)]
19. Dass, A.; Srivastava, S.; Chaudhary, G. Air pollution: A review and analysis using fuzzy techniques in Indian scenario. *Environ. Technol. Innov.* **2021**, *22*, 101441. [[CrossRef](#)]
20. Liang, C.S.; Duan, F.K.; He, K.B.; Ma, Y.L. Review on recent progress in observations, source identifications and countermeasures of PM_{2.5}. *Environ. Int.* **2016**, *86*, 150–170. [[CrossRef](#)]
21. Chen, Y.-C.; Chiang, H.-C.; Hsu, C.-Y.; Yang, T.-T.; Lin, T.-Y.; Chen, M.-J.; Chen, N.-T.; Wu, Y.-S. Ambient PM_{2.5}-bound polycyclic aromatic hydrocarbons (PAHs) in Changhua County, central Taiwan: Seasonal variation, source apportionment and cancer risk assessment. *Environ. Pollut.* **2016**, *218*, 372–382. [[CrossRef](#)]
22. Hopke, P.K.; Dai, Q.; Li, L.; Feng, Y. Global review of recent source apportionments for airborne particulate matter. *Sci. Total Environ.* **2020**, *740*, 140091. [[CrossRef](#)]
23. Han, F.; Kota, S.H.; Wang, Y.; Zhang, H. Source apportionment of PM_{2.5} in Baton Rouge, Louisiana during 2009–2014. *Sci. Total Environ.* **2017**, *586*, 115–126. [[CrossRef](#)]
24. Zhang, X.; Feng, X.; Tian, J.; Zhang, Y.; Li, Z.; Wang, Q.; Cao, J.; Wang, J. Dynamic harmonization of source-oriented and receptor models for source apportionment. *Sci. Total Environ.* **2023**, *859 Pt 1*, 160312. [[CrossRef](#)]
25. Moreda-Pineiro, J.; Turnes-Carou, I.; Alonso-Rodriguez, E.; Moscoso-Perez, C.; Blanco-Heras, G.; Lopez-Mahia, P.; Muniategui-Lorenzo, S.; Prada-Rodriguez, D. The influence of oceanic air masses on concentration of major ions and trace metals in PM_{2.5} fraction at a coastal European suburban site. *Water Air Soil Pollut.* **2015**, *226*, 2240. [[CrossRef](#)]
26. Ratanavalachai, T.; Trivitayanurak, W. Application of a PM_{2.5} dispersion model in the Bangkok central business district for air quality management. *Front. Environ. Sci. Sect. Atmos. Clim.* **2023**, *11*, 1237366. [[CrossRef](#)]
27. Fakhri, N.; Stevens, R.; Downey, A.; Oikonomou, K.; Sciare, J.; Afif, C.; Hayes, P. Source apportionment of PM_{2.5} in Montréal, Canada, and health risk assessment for potentially toxic elements. *Atmos. Chem. Phys.* **2024**, *24*, 1193–1212. [[CrossRef](#)]
28. Yin, X.; Huang, Z.; Zheng, J.; Yuan, Z.; Zhuy, W.; Huang, X.; Chen, D. Source contributions to PM_{2.5} in Guangdong province, China by numerical modeling: Results and implications. *Atmos. Res.* **2017**, *186*, 63–71. [[CrossRef](#)]
29. Liu, J.; Li, J.; Zhang, Y.; Liu, D.; Ding, P.; Shen, C.; Shen, K.; He, Q.; Ding, X.; Wang, X.; et al. Source apportionment using radiocarbon and organic tracers for PM_{2.5} carbonaceous aerosols in Guangzhou, South China: Contrasting local- and regional-scale haze events. *Environ. Sci. Technol.* **2014**, *48*, 12002–12011. [[CrossRef](#)]
30. Liao, K.; Zhang, J.; Chen, Y.; Lu, X.; Fung, J.C.H.; Ying, Q.; Yu, J.Z. Regional source apportionment of trace metals in fine particulate matter using an observation-constrained hybrid model. *NPJ Clim. Atmos. Sci.* **2023**, *6*, 65. [[CrossRef](#)]

31. Dhyni, R.; Sharma, N.; Maity, A.K. Prediction of PM_{2.5} along urban highway corridor under mixed traffic conditions using CALINE4 model. *J. Environ. Manag.* **2017**, *198*, 24–32. [\[CrossRef\]](#)
32. Di, Q.; Koutrakis, P.; Schwartx, J. A hybrid prediction model for PM_{2.5} mass and components using a chemical transport model and land use regression. *Atmos. Environ.* **2016**, *131*, 390–399. [\[CrossRef\]](#)
33. Yin, Q.; Wang, J.; Hu, M.; Wong, H. Estimation of daily PM_{2.5} concentration and its relationship with meteorological conditions in Beijing. *J. Environ. Sci.* **2016**, *48*, 161–168. [\[CrossRef\]](#)
34. Moisan, S.; Herrera, R.; Clements, A. A dynamic multiple equation approach for forecasting PM_{2.5} pollution in Santiago, Chile. *Int. J. Forecast.* **2018**, *34*, 566–581. [\[CrossRef\]](#)
35. Sun, W.; Li, Z. Hourly PM_{2.5} concentration forecasting based on mode decomposition-recombination technique and ensemble learning approach in severe haze episodes of China. *J. Clean. Prod.* **2020**, *263*, 121442. [\[CrossRef\]](#)
36. Ni, X.Y.; Huang, H.; Du, W.P. Relevance analysis and short-term prediction of PM_{2.5} concentrations in Beijing based on multi-source data. *Atmos. Environ.* **2017**, *150*, 146–161. [\[CrossRef\]](#)
37. Zhao, L.; Li, Z.; Qu, L. Forecasting of Beijing PM_{2.5} with a hybrid ARIMA model based on integrated AIC and improved GS fixed-order methods and seasonal decomposition. *Heliyon* **2022**, *8*, e12239. [\[CrossRef\]](#)
38. Mogollón-Sotelo, C.; Casallas, A.; Vidal, S. A support vector machine model to forecast ground-level PM_{2.5} in a highly populated city with a complex terrain. *Air Qual. Atmos. Health* **2021**, *14*, 399–409. [\[CrossRef\]](#)
39. Di, Q.; Amini, H.; Shi, L.; Kloog, I.; Silvern, R.; Kelly, J.; Sabath, M.B.; Choirat, C.; Koutrakis, P.; Lyapusting, A.; et al. An ensemble-based model of PM_{2.5} concentration across the contiguous United States with high spatiotemporal resolution. *Environ. Int.* **2019**, *130*, 104909. [\[CrossRef\]](#) [\[PubMed\]](#)
40. Xiao, F.; Yang, M.; Fan, H.; Fan, G.; Al-qaness, M.A.A. An improved deep learning model for predicting daily PM_{2.5} concentration. *Sci. Rep.* **2020**, *10*, 20988. [\[CrossRef\]](#)
41. Qin, D.; Yun, J.; Zou, G.; Yong, R.; Zhao, Q.; Zhang, B. A novel combined prediction scheme based on CNN and LSTM for urban PM_{2.5} concentration. *IEEE Access* **2019**, *7*, 20050–20059. [\[CrossRef\]](#)
42. Ausati, S.; Amanollahi, J. Assessing the accuracy of ANFIS, EEMD-GRNN, PCR, and MLR models in predicting PM_{2.5}. *Atmos. Environ.* **2016**, *142*, 465–474. [\[CrossRef\]](#)
43. Martinez-Zeron, E.; Aceves-Fernandez, M.A.; Gorrostieta-Hurtado, E.; Sotomayor-Olmedo, A.; Ramos-Arreguín, J.M. Method to improve airborne pollution forecasting by using ant colony optimization and neuro-fuzzy algorithms. *Int. J. Intell. Sci.* **2014**, *4*, 81–90. [\[CrossRef\]](#)
44. Wang, P.; Zhang, H.; Qin, Z.; Zhang, G. A novel hybrid-Garch model based on ARIMA and SVM for PM_{2.5} concentrations forecasting. *Atmos. Pollut. Res.* **2017**, *8*, 850–860. [\[CrossRef\]](#)
45. Zhang, B.; Li, X.; Zhao, Y.; Li, Y.; Wang, X. Air quality PM_{2.5} prediction based on multi-model fusion. In Proceedings of the 2019 Chinese Control and Decision Conference (CCDC), Nanchang, China, 3–5 June 2019.
46. Cai, P.; Zhang, C.; Chai, J. Forecasting hourly PM_{2.5} concentration based on decomposition-ensemble-reconstruction framework incorporating deep learning algorithms. *Data Sci. Manag.* **2023**, *6*, 46–54. [\[CrossRef\]](#)
47. Lee, W.C.; Shen, L.; Catalano, P.J.; Mickley, L.J.; Koutrakis, P. Effect of future temperature change on PM_{2.5} infiltration in the greater Boston area. *Atmos. Environ.* **2017**, *150*, 98–105. [\[CrossRef\]](#)
48. Li, L.; Fu, Y.; Fung, J.; Tse, K.; Lau, A. Development of a back-propagation neural network combined with an adaptive multi-objective particle swarm optimizer algorithm for predicting and optimizing indoor CO₂ and PM_{2.5} concentrations. *J. Build. Eng.* **2022**, *54*, 104600. [\[CrossRef\]](#)
49. Zhou, Y.; Yang, G. A predictive model of indoor PM_{2.5} considering occupancy level in a hospital outpatient hall. *Sci. Total Environ.* **2022**, *844*, 157233. [\[CrossRef\]](#)
50. Li, Z.; Tong, X.; Ho, J.; Kwok, T.; Dong, G.; Ho, K.; Yim, S. A practical framework for predicting residential indoor PM_{2.5} concentration using land-use regression and machine learning methods. *Chemosphere* **2021**, *265*, 129140. [\[CrossRef\]](#)
51. Elbayoumi, M.; Ramli, N.; Yusof, N. Development and comparison of regression models and feedforward backpropagation neural network models to predict seasonal indoor PM_{2.5–10} and PM_{2.5} concentrations in naturally ventilated schools. *Atmos. Pollut. Res.* **2015**, *6*, 1013–1023. [\[CrossRef\]](#)
52. Wang, J.; Han, J.; Li, T.; Wu, T.; Fang, C. Impact analysis of meteorological variables on PM_{2.5} pollution in the most polluted cities in China. *Heliyon* **2023**, *9*, e17609. [\[CrossRef\]](#)
53. Hsu, C.H.; Cheng, F.Y. Classification of weather patterns to study the influence of meteorological characteristics on PM_{2.5} concentrations in Yunlin County, Taiwan. *Atmos. Environ.* **2016**, *144*, 397–408. [\[CrossRef\]](#)
54. Yin, P.-Y. A novel spatiotemporal analysis framework for air pollution episode association in Puli, Taiwan. *Appl. Sci.* **2023**, *13*, 5808. [\[CrossRef\]](#)
55. Tsai, I.C.; Hsieh, P.R.; Hsu, H.H.; Tung, Y.S.; Chen, Y.M.; Cheng, C.T. Climate change-induced impacts on PM_{2.5} in Taiwan under 2 and 4 °C global warming. *Atmos. Pollut. Res.* **2024**, *15*, 102106. [\[CrossRef\]](#)

56. Danek, T.; Węglińska, E.; Zareba, M. The influence of meteorological factors and terrain on air pollution concentration and migration: A geostatistical case study from Krakow, Poland. *Sci. Rep.* **2022**, *12*, 11050. [\[CrossRef\]](#) [\[PubMed\]](#)
57. Reikard, G. Volcanic emissions and air pollution: Forecasts from time series models. *Atmos. Environ. X* **2019**, *1*, 100001. [\[CrossRef\]](#)
58. Carreras-Sospedra, M.; Zhu, S.; MacKinnon, M.; Lassman, W.; Mirocha, J.D.; Barbato, M.; Dabdub, D. Air quality and health impacts of the 2020 wildfires in California. *Fire Ecol.* **2024**, *20*, 6. [\[CrossRef\]](#)
59. Tan, P.H.; Chou, C.; Chou, C.C.K. Impact of urbanization on the air pollution “holiday effect” in Taiwan. *Atmos. Environ.* **2013**, *70*, 361–375. [\[CrossRef\]](#)
60. Zhao, S.; Yu, Y.; Yin, D.; Liu, N.; He, J. Ambient particulate pollution during Chinese Spring Festival in urban Lanzhou, Northwestern China. *Atmos. Pollut. Res.* **2014**, *5*, 335–343. [\[CrossRef\]](#)
61. Yin, P.Y. Mining associations between air quality and natural and anthropogenic factors. *Sustainability* **2024**, *16*, 4614. [\[CrossRef\]](#)
62. Li, S.; Li, Z.; Ni, J.; Yuan, J. Growing pains for others: Using holidays to identify the pollution spillover between China and South Korea. *China Econ. Rev.* **2023**, *77*, 01916. [\[CrossRef\]](#)
63. Singh, V.; Sokhi, R.S.; Kukkonen, J. An approach to predict population exposure to ambient air PM_{2.5} concentrations and its dependence on population activity for the megacity London. *Environ. Pollut.* **2020**, *257*, 113623. [\[CrossRef\]](#)
64. Yousefi, R.; Shaheen, A.; Wang, F.; Ge, Q.; Wu, R.; Lelieveld, J.; Wang, J.; Su, X. Fine particulate matter (PM_{2.5}) trends from land surface changes and air pollution policies in China during 1980–2020. *J. Environ. Manag. Part B* **2023**, *326*, 116847. [\[CrossRef\]](#)
65. Zhao, L.; Zhang, M.; Cheng, S.; Fang, Y.; Wang, S.; Zhou, C. Investigate the effects of urban land use on PM_{2.5} concentration: An application of deep learning simulation. *Build. Environ.* **2023**, *242*, 110521. [\[CrossRef\]](#)
66. Milicevic, O.; Salom, I.; Rodic, A.; Markovic, S.; Tumbas, M.; Zigic, D.; Djordjevic, M.; Djordjevic, M. PM_{2.5} as a major predictor of COVID-19 basic reproduction number in the USA. *Environ. Res.* **2021**, *201*, 111526. [\[CrossRef\]](#) [\[PubMed\]](#)
67. Khatibi, T.; Karampour, N. Predicting the number of hospital admissions due to mental disorders from air pollutants and weather condition descriptors using stacked ensemble of Deep Convolutional models and LSTM models (SEDCMLM). *J. Clean. Prod.* **2021**, *280 Pt 1*, 124410. [\[CrossRef\]](#)
68. Wan, S.; Zhao, X.; Niu, Z.; Dong, L.; Wu, Y.; Gu, S.; Feng, Y.; Hua, X. Influence of ambient air pollution on successful pregnancy with frozen embryo transfer: A machine learning prediction model. *Ecotoxicol. Environ. Saf.* **2022**, *236*, 113444. [\[CrossRef\]](#)
69. Fan, J.; Wu, L.; Zhang, F.; Cai, H.; Wang, X.; Lu, X.; Xiang, Y. Evaluating the effect of air pollution on global and diffuse solar radiation prediction using support vector machine modeling based on sunshine duration and air temperature. *Renew. Sustain. Energy Rev.* **2018**, *94*, 732–747. [\[CrossRef\]](#)
70. Vakili, M.; Sabbagh-Yazdi, S.R.; Kalhor, K.; Khosrojerdi, S. Using Artificial Neural Networks for Prediction of Global Solar Radiation in Tehran Considering Particulate Matter Air Pollution. *Energy Procedia* **2015**, *74*, 1205–1212. [\[CrossRef\]](#)
71. Song, Z.; Wang, M.; Yang, H. Quantification of the impact of fine particulate matter on solar energy resources and energy performance of different photovoltaic technologies. *ACS Environ. Au* **2022**, *2*, 275–286. [\[CrossRef\]](#)
72. Munir, S.; Gabr, S.; Habeebullah, T.M.; Janajrah, M.A. Spatiotemporal analysis of fine particulate matter (PM_{2.5}) in Saudi Arabia using remote sensing data. *Egypt. J. Remote Sens. Space Sci.* **2016**, *19*, 195–205. [\[CrossRef\]](#)
73. Bai, K.; Ma, M.; Chang, N.B.; Gao, W. Spatiotemporal trend analysis for fine particulate matter concentrations in China using high-resolution satellite-derived and ground-measured PM_{2.5} data. *J. Environ. Manag.* **2019**, *233*, 530–542. [\[CrossRef\]](#)
74. Bai, Y.; Liu, M. Multi-scale spatiotemporal trends and corresponding disparities of PM_{2.5} exposure in China. *Environ. Pollut.* **2024**, *340 Pt 2*, 122857. [\[CrossRef\]](#)
75. Yue, H.; Huang, Q.; He, C.; Zhang, X.; Fang, Z. Spatiotemporal patterns of global air pollution: A multi-scale landscape analysis based on dust and sea-salt removed PM_{2.5} data. *J. Clean. Prod.* **2020**, *252*, 119887. [\[CrossRef\]](#)
76. Saez, M.; Barceló, M.A. Spatial prediction of air pollution levels using a hierarchical Bayesian spatiotemporal model in Catalonia, Spain. *Environ. Model. Softw.* **2022**, *151*, 105369. [\[CrossRef\]](#)
77. Zareba, M.; Dlugosz, H.; Danek, T.; Węglińska, E. Big-data-driven machine learning for enhancing spatiotemporal air pollution pattern analysis. *Atmosphere* **2023**, *14*, 760. [\[CrossRef\]](#)
78. Rincon, G.; Morantes, G.; Roa-López, H.; Cornejo-Rodriguez, M.d.P.; Jones, B.; Cremades, L.V. Spatio-temporal statistical analysis of PM₁ and PM_{2.5} concentrations and their key influencing factors at Guayaquil city, Ecuador. *Stoch. Environ. Res. Risk Assess.* **2023**, *37*, 1093–1117. [\[CrossRef\]](#)
79. Di, Q.; Kloog, I.; Koutrakis, P.; Lyapustin, A.; Wang, Y.; Schwartz, J. Assessing PM_{2.5} Exposures with High Spatiotemporal Resolution across the Continental United States. *Environ. Sci. Technol.* **2016**, *50*, 4712–4721. [\[CrossRef\]](#)
80. Yin, P.Y. Spatiotemporal retrieval and feature analysis of air pollution episodes. *Math. Biosci. Eng.* **2023**, *20*, 16824–16845. [\[CrossRef\]](#)
81. Parajapati, A.; Bechtel, J.; Ganesan, S. Condition based maintenance: A survey. *J. Qual. Maint. Eng.* **2012**, *18*, 384–400. [\[CrossRef\]](#)
82. Basri, E.I.; Abdul Razak, I.H.; Ab-Samat, H.; Kamaruddin, S. Preventive maintenance (PM) planning: A review. *J. Qual. Maint. Eng.* **2017**, *23*, 114–143. [\[CrossRef\]](#)

83. Feng, T.; Sun, Y.; Shi, Y.; Ma, J.; Feng, C.; Chen, Z. Air pollution control policies and impacts: A review. *Renew. Sustain. Energy Rev.* **2024**, *191*, 114071. [[CrossRef](#)]
84. Kumar, A.; Patil, R.S.; Dikshit, A.K.; Islam, S.; Kumar, R. Evaluation of control strategies for industrial air pollution sources using American Meteorological Society/Environmental Protection Agency Regulatory Model with simulated meteorology by Weather Research and Forecasting Model. *J. Clean. Prod.* **2016**, *116*, 110–117. [[CrossRef](#)]
85. Zareba, M.; Cogiel, S.; Danek, T.; Weglinska, E. Machine Learning Techniques for Spatio-Temporal Air Pollution Prediction to Drive Sustainable Urban Development in the Era of Energy and Data Transformation. *Energies* **2024**, *17*, 2738. [[CrossRef](#)]
86. Shi, H.; Fan, J.; Zhao, D. Predicting household PM_{2.5}-reduction behavior in Chinese urban areas: An integrative model of theory of planned behavior and norm activation theory. *J. Clean. Prod.* **2017**, *145*, 64–73. [[CrossRef](#)]
87. Shi, H.; Wang, S.; Guo, S. Predicting the impacts of psychological factors and policy factors on individual's PM_{2.5} reduction behavior: An empirical study in China. *J. Clean. Prod.* **2019**, *241*, 118416. [[CrossRef](#)]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.